

Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

WARNING

Depending on the circumstance, fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the battery cables. See equipment manufacturer service information.
- Remove the fuel filters. Refer to Procedure 006-015 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-015-tr.html)

with Electronically Actuated Injector

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- Disconnect the battery cables. See equipment manufacturer service information.
- Disconnect the fuel temperature sensor. Use the following procedure found in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair,

Bulletin 4021533. Refer to Procedure 019-119 in Section 19.

(/qs3/pubsys2/xml/en/procedures/05/05-019-119.html)

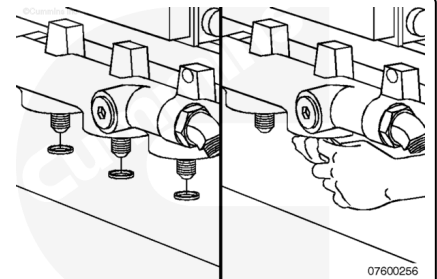
- Disconnect the fuel pressure sensor. Use the following procedure found in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-398 in Section 19.
(/qs3/pubsys2/xml/en/procedures/05/05-019-398.html)
- Remove the fuel filters. Refer to Procedure 006-015 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-015-tr.html)
- Disconnect the fuel supply lines. Refer to Procedure 006-024 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-024-tr.html)

Remove

with Mechanically Actuated Injector

Remove the fuel filter supply and return hoses.

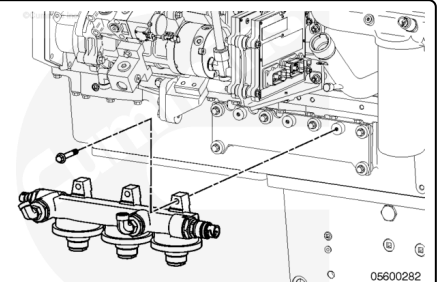
Remove the fuel filter head mounting capscrews and the fuel filter head.



with Electronically Actuated Injector

Remove the six fuel filter head mounting capscrews and the fuel filter head.

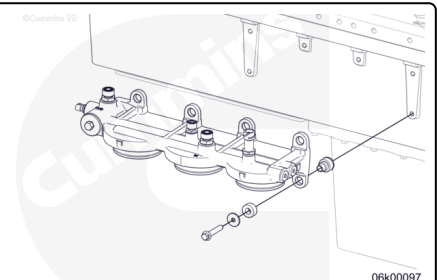
Note : For marine applications, the location of the fuel filter head can vary.



For Generator Model DQKAN with Low NOx Emissions Capability

Note : Some graphics in this procedure are generic and do not depict the correct number of fuel filters.

Remove fuel filter head mounting capscrews and the fuel filter head.



Disassemble

with Mechanically Actuated Injector

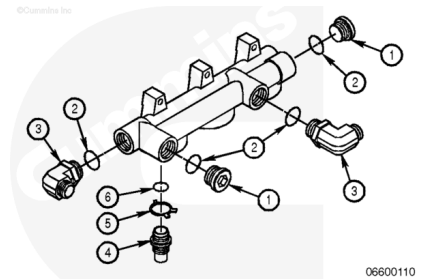
Remove the straight thread o-ring plugs (1) and the o-rings (2).

Remove the inlet and outlet elbows (3) and the o-rings (2).

The fuel filter adapters (4) do **not** need to be removed unless they are loose or damaged.

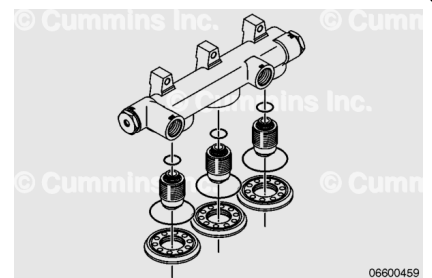
If it is necessary to remove the fuel adapter, remove the adapters (4), lock plates (5), and o-rings (6).

Discard all removed o-rings.



Remove the filter head adapter spuds and the adapter rings.

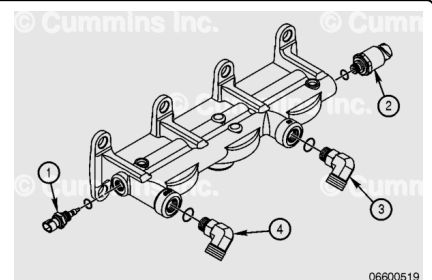
Remove and discard the six o-rings.



with Electronically Actuated Injector

Stage Two Filter Head

- Remove the fuel temperature sensor (1). Use the following procedure found in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-398 in Section 19. (</qs3/pubsys2/xml/en/procedures/05/05-019-398.html>)
- Remove the fuel pressure sensor (2). Use the following procedure found in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-119 in Section 19. (</qs3/pubsys2/xml/en/procedures/05/05-019-119.html>)
- Remove the fuel inlet (3) and fuel outlet (4) fittings.
- Discard the o-rings.



Note : Plugs can be used in locations (1) and (2) instead of sensors, if required.

For Generator Model DQKAN with Low NOx Emissions Capability

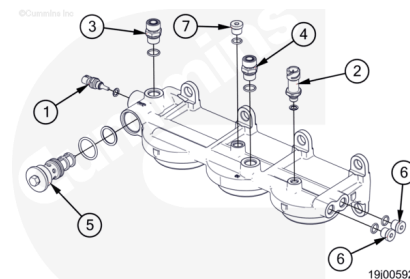
Remove the fuel temperature sensor (1). Use the following procedure in the QSK38, QSK50, and QSK60 CM2150 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4022102. Refer to Procedure 019-119 (</qs3/pubsys2/xml/en/procedures/122/122-019-119.html>) in Section 19.

Remove the fuel pressure sensor (2). Use the following procedure in the QSK38, QSK50, and QSK60 CM2150 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4022102. Refer to Procedure 019-398 (</qs3/pubsys2/xml/en/procedures/122/122-019-398.html>) in Section 19.

Remove the fuel outlet (3), and fuel inlet (4). Discard o-rings.

Remove the filter screen assembly (5) and discard the o-rings.

Remove the threaded plugs (6 and 7) and discard the o-rings.



Clean and Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

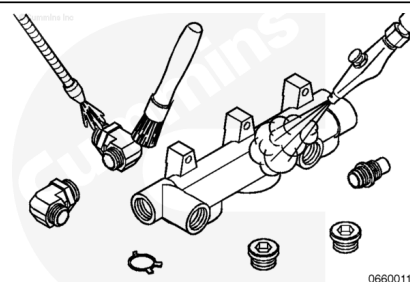
When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

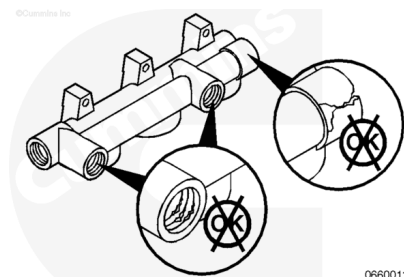
Use a solvent that will not damage aluminum or the remaining o-rings.



Use solvent to clean the filter head components.

Use compressed air to dry the filter head components.

Inspect the filter head for damaged threads or cracks in the housing.

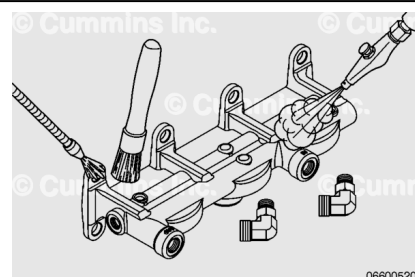


06600112

with Electronically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



06600520

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

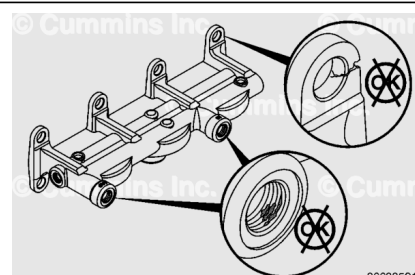
⚠ CAUTION ⚠

Use a solvent that will not damage aluminum or the remaining o-rings.

Stage Two Filter Head

- Use solvent to clean the filter head components.
- Use compressed air to dry the filter head components.

Inspect the filter head for damaged threads or cracks in the housing.

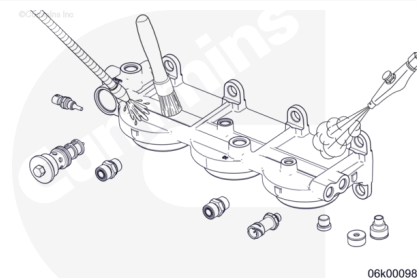


06600521

For Generator Model DQKAN with Low NOx Emissions Capability:

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

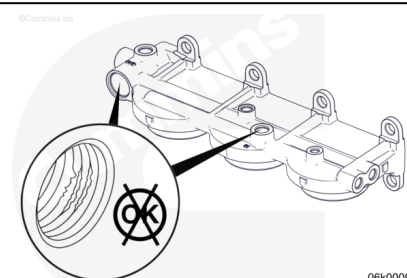
Use a solvent that will not damage aluminum or the remaining o-rings.

Clean the filter head components with solvent and dry with compressed air.

Inspect filter head for damaged threads or cracks in the housing.

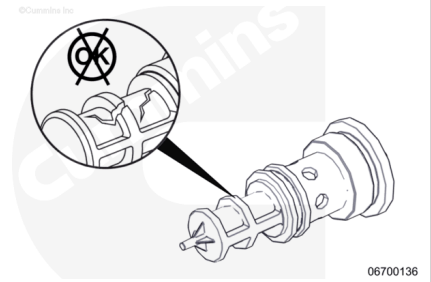
Inspect the fuel pressure sensor. Use the following procedure in the QSK38, QSK50, and QSK60 CM2150 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4022102. Refer to Procedure 019-398 (</qs3/pubsys2/xml/en/procedures/122/122-019-398.html>) in Section 19.

Inspect the fuel temperature sensor. Use the following procedure in the QSK38, QSK50, and QSK60 CM2150 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4022102. Refer to Procedure 019-119 (</qs3/pubsys2/xml/en/procedures/122/122-019-119.html>) in Section 19.

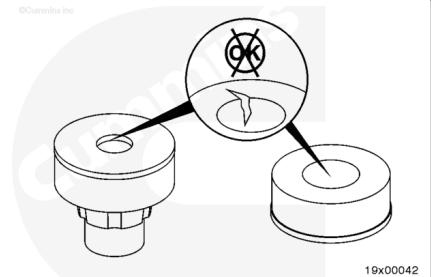


Inspect the filter screen assembly for cracks, tears, and debris.

If cracks, tears, or debris are visible, the filter screen assembly **must** be replaced.



Inspect the vibration isolators for damage. Replace if damaged.



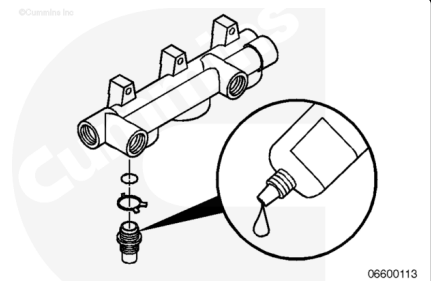
Assemble

with Mechanically Actuated Injector

Install new o-rings.

Install the lock plate. Locate the 90-degree bend in the drilled hole in the fuel filter head.

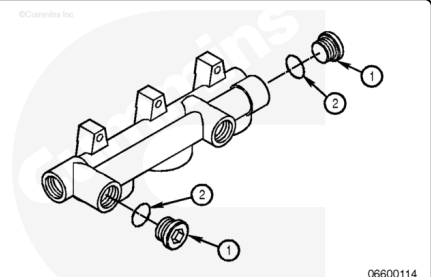
Apply one drop of Loctite® 601 on the threads of the adapter that engages the threads in the filter head and install the adapter.



Torque Value: 102 n•m [75 ft-lb]

Note : The position of the plugs and the inlet and outlet elbows can be different than illustrated due to the various fuel plumbing installations.

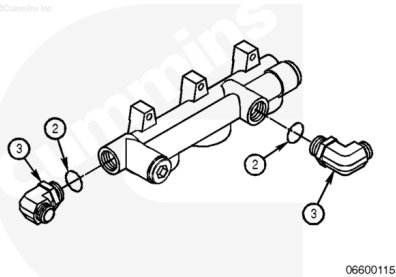
Use new o-rings (2) to install the straight-thread o-ring plugs (1).



Torque Value: 102 n•m [75 ft-lb]

Use new o-rings (2) to install the fuel inlet and outlet elbows (3).

Torque Value: 102 n•m [75 ft-lb]



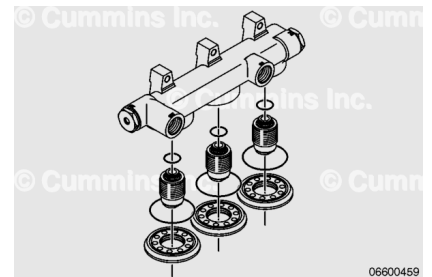
Install new o-rings above the filter head adapter spuds. Apply thread locker to the smaller threaded portion of the spud.

Install the spuds in the filter head.

Torque Value: 88 n•m [65 ft-lb]

Install new o-rings above the filter head adapter rings. Apply thread locking compound to the inside of the threads. Thread the rings into the spuds until the o-rings contact the filter head. Use care to make sure the o-ring is properly seated.

Torque Value: 88 n•m [65 ft-lb]

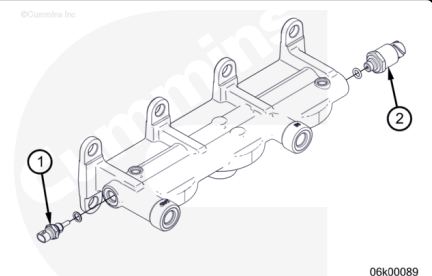


with Electronically Actuated Injector

- Install the fuel temperature sensor (1). Use the following procedure found in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-119 in Section 19. (</qs3/pubsys2/xml/en/procedures/05/05-019-119.html>)
- Install the fuel pressure sensor (2). Use the following procedure found in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-398 in Section 19. (</qs3/pubsys2/xml/en/procedures/05/05-019-398.html>)

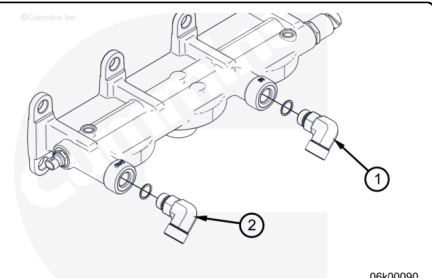
M18 straight threaded plugs can be used in locations (1) and (2) instead of sensors, if required.

Torque Value: 41 n•m [30 ft-lb]



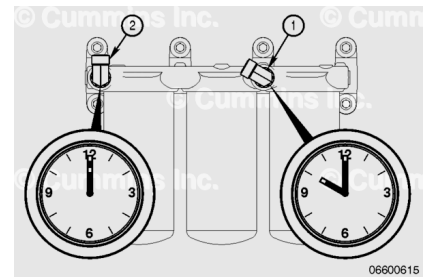
Use new o-rings and install the fuel inlet (1) and fuel outlet (2) fittings.

Torque Value: 62 n•m [46 ft-lb]



Make sure the fuel outlet and inlet elbow joints are positioned at 10 o'clock (1) and 12 o'clock (2) orientations as shown.

Torque Value: 115 n•m [85 ft-lb]

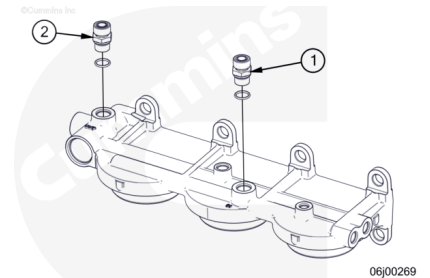


For Generator Model DQKAN with Low NOx Emissions Capability

Install the union fittings into the filter head at the following locations.

1. Fuel inlet
2. Fuel outlet

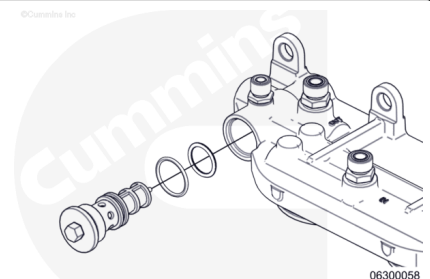
Torque Value: 85 n•m [63 ft-lb]



Install new o-rings onto the filter screen assembly.

Install the filter screen assembly into the fuel filter head.

Torque Value: 60 n•m [44 ft-lb]

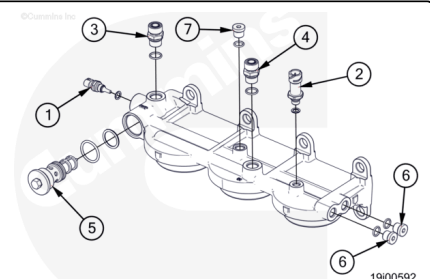


Install the fuel temperature sensor (1). Use the following procedure in the QSK38, QSK50, and QSK60 CM2150 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4022102. Refer to Procedure 019-119 (</qs3/pubsys2/xml/en/procedures/122/122-019-119.html>) in Section 19.

Install the fuel pressure sensor (2). Use the following procedure in the QSK38, QSK50, and QSK60 CM2150 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4022102. Refer to Procedure 019-398 (</qs3/pubsys2/xml/en/procedures/122/122-019-398.html>) in Section 19.

Install new o-rings on the threaded plugs (6) and install the plugs.

Torque Value: 40 n•m [26 ft-lb]



Install new o-ring on threaded plug (7) and install plug.

Torque Value: 35 n•m [26 ft-lb]

Install

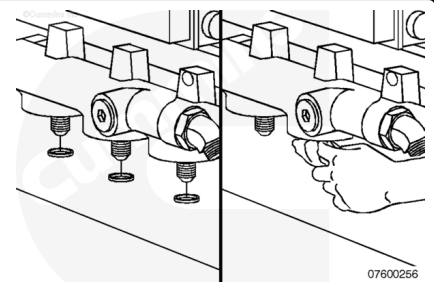
with Mechanically Actuated Injector

Install the fuel filter head and install the fuel filter head mounting capscrews.

Torque Value: 55 n•m [41 ft-lb]

Install the fuel filter supply and return hoses.

Clean the gasket sealing surface with a clean, lint-free towel.



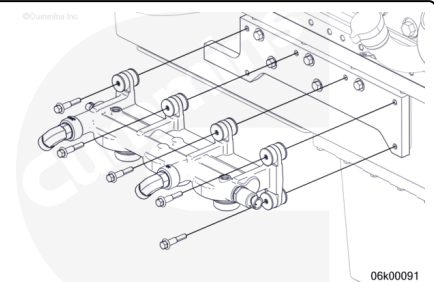
with Electronically Actuated Injector

Stage 2 Filter Head

- Install the fuel filter head assembly to the engine with the six mounting capscrews.

Torque Value: 45 n•m [33 ft-lb]

- Install the fuel supply and return hoses.
- Connect the engine wiring harness to the pressure and temperature sensors on the fuel filter head, if equipped.

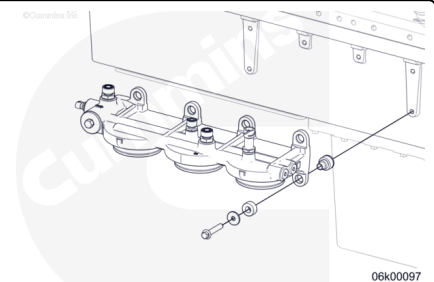


For Generator Model DQKAN with Low NOx Emissions Capability

Note : If the fuel filter head is remote-mounted, reference the equipment manufacturer service information.

Install the fuel filter head assembly onto the engine with the two-piece vibration isolators and six capscrews.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

with Mechanically Actuated Injector

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the fuel filters. Refer to Procedure 006-015 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-015-tr.html)
- Connect the battery cables. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/56/56-013-009.html)

with Electronically Actuated Injector

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the fuel supply lines. Refer to Procedure 006-024 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-024-tr.html)
- Install the fuel filters. Refer to Procedure 006-015 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-015-tr.html)
- Connect the fuel temperature sensor. Use the following procedure found in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-119 in Section 19.
(/qs3/pubsys2/xml/en/procedures/05/05-019-119.html)
- Connect the fuel pressure sensor. Use the following procedure found in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Troubleshooting and Repair, Bulletin 4021533. Refer to Procedure 019-398 in Section 19.
(/qs3/pubsys2/xml/en/procedures/05/05-019-398.html)
- Connect the battery cables. See equipment manufacturer service information.
- Operate the engine and check for leaks.

For Generator Model DQKAN with Low NOx Emissions Capability

WARNING

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- Install fuel supply lines. Refer to Procedure 006-024 (</qs3/pubsys2/xml/en/procedures/56/56-006-024-tr.html>) in Section 6.
- Install fuel filter(s). Refer to Procedure 006-076 (</qs3/pubsys2/xml/en/procedures/56/56-006-076-tr.html>) in Section 6.
- Connect fuel temperature sensor. Use the following procedure in the QSK38, QSK50, and QSK60 CM2150 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4022102. Refer to Procedure 019-119 (</qs3/pubsys2/xml/en/procedures/122/122-019-119.html>) in Section 19.
- Connect fuel pressure sensor. Use the following procedure in the QSK38, QSK50, and QSK60 CM2150 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4022102. Refer to Procedure 019-398 (</qs3/pubsys2/xml/en/procedures/122/122-019-398.html>) in Section 19.
- Connect batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.